MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.6: Directional derivatives, critical points, and gradient descent

---

© 2024–2026 Jacob Denson, Kaitlyn Phillipson, Sebastien Roch. Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Directional derivatives

By Linear Approximation, the first-order change in a scalar-valued function near $\mathbf{a}$ is

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \approx \nabla f(\mathbf{a})^T \mathbf{h}.$$

To study change along a unit direction $\mathbf{v}$, restrict the input to the line

$$\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}.$$

The resulting one-variable rate of change is the directional derivative. The chain rule for one-parameter inputs in Chain Rule for One Independent Variable gives the gradient formula below.

Definition

Let f be a scalar-valued function. Then if $\mathbf{v}$ is a unit vector, the directional derivative of f in the direction of the vector $\mathbf{v}$, denoted $D_{\mathbf{v}} f$, is defined as

$$D_{\mathbf{v}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a}+h\mathbf{v}) - f(\mathbf{a})}{h},$$

provided the limit exists.

The directional derivative represents the rate of change of the function f at the point $\mathbf{a}$ in the direction of the unit vector $\mathbf{v}$. For a function $f(x, y)$ of two variables, the directional derivative $D_{\mathbf{v}} f(\mathbf{a})$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane containing both the line passing through $\mathbf{a}$ with direction vector $\mathbf{v}$, and the z axis.

Remark

Partial derivatives are a special case of directional derivatives, i.e., we have

$$\frac{\partial f}{\partial x_i} = D_{\mathbf{e}_i} f.$$

Despite partial derivatives being a special case of directional derivatives, in most circumstances we can compute directional derivatives using partial derivatives, or equivalently, the gradient.

Theorem

Suppose f is a function differentiable at a point $\mathbf{a}$. Then for each unit vector $\mathbf{v}$, $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

Why is this true?

Define the path $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$ and the single-variable function $g(t) = f(\mathbf{c}(t)) = f(\mathbf{a} + t\mathbf{v})$. Since $\mathbf{c}(0) = \mathbf{a}$, we have $g(0) = f(\mathbf{a})$. Therefore, the definition of the directional derivative gives

$$\begin{aligned} D_{\mathbf{v}} f(\mathbf{a}) &= \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h} \\ &= \lim_{h \to 0} \frac{g(h) - g(0)}{h} \\ &= g'(0). \end{aligned}$$

Write $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$. The ith coordinate of $\mathbf{c}(t)$ is $c_i(t) = a_i + tv_i$, so $c_i'(0) = v_i$. Because $g = f \circ \mathbf{c}$, the chain rule (Chain Rule for One Independent Variable) gives

$$\begin{aligned} g'(0) &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{c}(0))c_i'(0) \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a})v_i \\ &= \nabla f(\mathbf{a}) \cdot \mathbf{v}. \end{aligned}$$

Combining the two calculations proves that $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

The theorem greatly simplifies the computations of directional derivatives.

Example: Directional derivatives and unit directions (U3-LO8)
Let $f(x, y) = x^2 y + y^2$.

Task
Compute the directional derivative $D_{\mathbf{v}} f(1, 2)$, where $\mathbf{v} = (1/\sqrt{2}, 1/\sqrt{2})$, and $f(x, y) = x^2 y + y^2$.

We begin by computing the gradient of f at $(1, 2)$. We have

$$f_x(x, y) = 2xy \quad \text{and} \quad f_y(x, y) = x^2 + 2y.$$

So $\nabla f(x, y) = (2xy, x^2 + 2y)$, and in particular

$$\nabla f(1,2) = \begin{bmatrix} 2(1)(2) \\ (1)^2 + 2(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

The given vector $\mathbf{v}$ is already a unit vector because

$$\|\mathbf{v}\| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1.$$

Therefore, we may apply the directional-derivative formula directly:

$$D_{\mathbf{v}} f(1,2) = \nabla f(1,2) \cdot \mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 9/\sqrt{2}.$$

Task

Find the directional derivative of f at the point $(-1, 3)$ in the direction of the vector $(-1, \sqrt{3})$. That is, compute $D_{\mathbf{v}} f(-1, 3)$, where $\mathbf{v}$ is the unit vector in the direction of $(-1, \sqrt{3})$.

The vector

$$\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$$

has norm

$$\sqrt{(-1)^2 + (\sqrt{3})^2} = 2.$$

So the unit vector in this direction is

$$\mathbf{v} = \frac{1}{2}\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}.$$

For

$$f(x, y) = x^2 y + y^2,$$

we have

$$\nabla f(x, y) = \begin{bmatrix} 2xy \\ x^2 + 2y \end{bmatrix}.$$

Thus

$$\nabla f(-1, 3) = \begin{bmatrix} -6 \\ 7 \end{bmatrix}.$$

Therefore

$$
\begin{aligned}
D_{\mathbf{v}} f(-1,3) &= \nabla f(-1,3) \cdot \mathbf{v} \\
&= \begin{bmatrix} -6 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} \\
&= 3 + \frac{7\sqrt{3}}{2} = \frac{6 + 7\sqrt{3}}{2}.
\end{aligned}
$$

Activity: A directional derivative between two points (U3-LO8)

Find the directional derivative of f at $(2, 1)$ in the direction from $(2, 1)$ to $(3, 5)$.

The unit vector pointing in the direction from $(2, 1)$ to $(3, 5)$ is the unit vector pointing in the same direction as the vector $(3 - 2, 5 - 1) = (1, 4)$. This vector has magnitude $\sqrt{1^2 + 4^2} = \sqrt{17}$, so the unit vector pointing in this direction is equal to

$$
\mathbf{v} = \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix}.
$$

Now

$$\nabla f(x, y) = \begin{bmatrix} 3x^2 y \\ x^3 \end{bmatrix},$$

and in particular,

$$\nabla f(2, 1) = \begin{bmatrix} 3(2)^2(1) \\ (2)^3 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}.$$

So

$$\begin{aligned} D_{\mathbf{v}} f(2, 1) &= \nabla f(2, 1) \cdot \mathbf{v} \\ &= \begin{bmatrix} 12 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix} \\ &= \frac{12 + 32}{\sqrt{17}} = \frac{44}{\sqrt{17}}. \end{aligned}$$

Activity: Directional change toward a specified point (U3-LO8)

Let $f(x, y) = x^2 e^y$. Find the directional derivative of f at $(2, 0)$ in the direction pointing from $(2, 0)$ to $(3, 0)$.

A vector pointing from $(2, 0)$ to $(3, 0)$ is

$$\mathbf{d} = \begin{bmatrix} 3-2 \\ 0-0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Its norm is

$$\|\mathbf{d}\| = \sqrt{1^2 + 0^2} = 1.$$

Thus this vector is already a unit vector. Therefore the required direction is

$$\mathbf{v} = \mathbf{d} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1.$$

Next,

$$\nabla f(x, y) = \begin{bmatrix} 2xe^y \\ x^2 e^y \end{bmatrix}, \qquad \nabla f(2, 0) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}.$$

Thus

$$
\begin{aligned}
D_{\mathbf{v}} f(2,0) &= \nabla f(2,0) \cdot \mathbf{v} \\
&= \begin{bmatrix} 4 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= 4 \cdot 1 + 4 \cdot 0 = 4.
\end{aligned}
$$

Activity: Alignment with the gradient (U3-LO8, U1-LO1)

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute $D_{\mathbf{v}} f(\mathbf{a})$ for each unit vector below:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_3 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{v}_4 = -\frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Which listed direction gives the greatest increase? Which gives the greatest decrease?

By the theorem,

$$D_{\mathbf{v}_1} f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3,$$

and

$$D_{\mathbf{v}_2} f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 4.$$

Also,

$$D_{\mathbf{v}_3} f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 5,$$

while

$$D_{\mathbf{v}_4} f(\mathbf{a}) = -5.$$

The greatest listed increase occurs in direction $\mathbf{v}_3$, which is aligned with the gradient. The greatest listed decrease occurs in direction $\mathbf{v}_4$, which points opposite the gradient.

Theorem

Suppose that a function f is differentiable at a point $\mathbf{a}$.

- If $\nabla f(\mathbf{a}) = 0$, then $D_{\mathbf{v}} f(\mathbf{a}) = 0$ for any unit vector $\mathbf{v}$.
- If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, then $D_{\mathbf{v}} f(\mathbf{a})$ is maximized, among unit vectors $\mathbf{v}$, when

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

It is minimized when

$$\mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

The maximum value is $\|\nabla f(\mathbf{a})\|$, and the minimum value is $-\|\nabla f(\mathbf{a})\|$.

Why is this true?

If $\nabla f(\mathbf{a}) = \mathbf{0}$, then the theorem gives

$$D_{\mathbf{v}} f(\mathbf{a}) = 0$$

for every unit vector $\mathbf{v}$.

Now suppose $\nabla f(\mathbf{a}) \neq \mathbf{0}$, and let θ be the angle between $\nabla f(\mathbf{a})$ and a unit vector $\mathbf{v}$. By the theorem and Angle between vectors,

$$D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| \cos\theta.$$

The largest possible value of $\cos\theta$ is 1, attained when $\mathbf{v}$ points in the direction of the gradient. The smallest possible value is -1, attained when $\mathbf{v}$ points in the opposite direction. Therefore the maximum and minimum directional derivatives are

$$\|\nabla f(\mathbf{a})\| \quad \text{and} \quad -\|\nabla f(\mathbf{a})\|,$$

attained at

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} \quad \text{and} \quad \mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|},$$

respectively.

Note: Level curves and steepest change

For a function of two variables with continuous partial derivatives and a nonzero gradient at $\mathbf{a}$, the theorem tells us that $\nabla f(\mathbf{a})$ is perpendicular to the level curve through $\mathbf{a}$. Moving along that curve leaves f unchanged, so a tangent direction gives no first-order change in f.

This gives a geometric explanation of the theorem. For motion at unit speed, any tangential component uses part of the motion without contributing to the directional derivative. The greatest increase therefore occurs when the motion is entirely perpendicular to the level curve, in the gradient direction; the greatest decrease occurs in the opposite direction. The dot-product formula identifies which way is uphill: a unit vector in the gradient direction gives the positive rate $\|\nabla f(\mathbf{a})\|$.

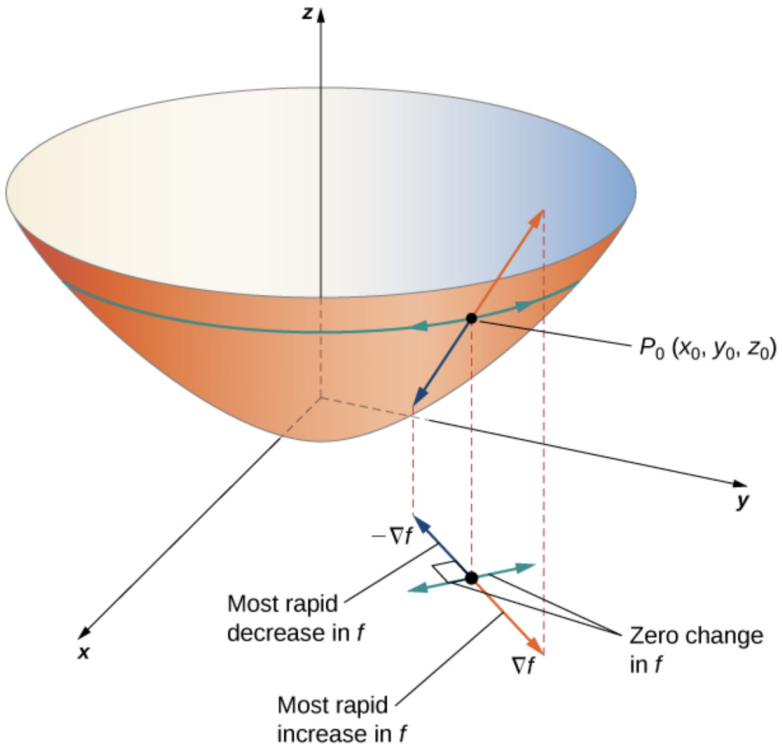


Figure. Gradient directions at a point on a surface.

Activity: Finding the direction of greatest increase (U3-LO8)

Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value of the directional derivative?

First, we calculate that

$$\nabla f(x, y) = \begin{bmatrix} 6x - 4y \\ -4x + 4y \end{bmatrix}.$$

So

$$\nabla f(-2, 3) = \begin{bmatrix} 6(-2) - 4(3) \\ -4(-2) + 4(3) \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix}.$$

the theorem tells us that the directional derivative $D_{\mathbf{v}} f$ is maximized when $\mathbf{v}$ is the unit vector pointing in the direction of $(-24, 20)$. Since $(-24, 20)$ has magnitude $\sqrt{(-24)^2 + (20)^2} = \sqrt{976}$, we see the direction which maximizes the directional derivative is

$$\mathbf{v} = \begin{bmatrix} -24/\sqrt{976} \\ 20/\sqrt{976} \end{bmatrix},$$

and the value of the directional derivative in this direction is equal to the magnitude of the gradient, which is

$$\sqrt{976}.$$

Activity: A directional derivative in three variables (U3-LO8)

Calculate $D_{\mathbf{u}} f(1, -2, 3)$ in the direction of $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ for the function

$$f(x, y, z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$

We compute that

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = \sqrt{9} = 3$$

So the unit vector in the direction of $\mathbf{v}$ is the vector

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = -\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$$

Using the formula for the directional derivative,

we find that

$$\begin{aligned} D_{\mathbf{u}} f(x,y,z) &= \nabla f(x,y,z) \cdot \mathbf{u} \\ &= \begin{bmatrix} 10x - 2y + 3z \\ -2x + 2y - 4z \\ -4y + 2z + 3x \end{bmatrix} \cdot \mathbf{u} \\ &= -\frac{10x - 2y + 3z}{3} \\ &\quad + 2\frac{-2x + 2y - 4z}{3} \\ &\quad + 2\frac{-4y + 2z + 3x}{3}. \end{aligned}$$

At the point $(1, -2, 3)$, we evaluate that

$$\begin{aligned} D_{\mathbf{u}} f(1,-2,3) &= (23)\left(-\frac{1}{3}\right) \\ &\quad + (-18)\left(\frac{2}{3}\right) + (17)\left(\frac{2}{3}\right) \\ &= \frac{-25}{3} \end{aligned}$$

In Orthogonal projection, the same dot product will be interpreted as a

signed projection coefficient of the gradient onto the chosen direction.

Optimization problems and critical points

Directional derivatives describe first-order change along a chosen direction. We now use them to identify candidates for local extrema and to motivate an iterative method for decreasing a function.
In this unit, we apply multivariate calculus to solve *optimization problems*.

Definition: Optimization problems

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a scalar-valued function, and consider a set $U \subseteq \mathbb{R}^n$. Then we can consider two optimization problems, in which we call f the objective function of the problem, and U the feasible region of the optimization problem:

- The minimization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *smallest* among all elements of U (so the function is *minimized* on U by $\mathbf{x}^*$).
- The maximization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *largest* among all elements of U (so the function is *maximized* on U by $\mathbf{x}^*$).

Definition: Minima and maxima

Suppose f is a scalar-valued function. A point $\mathbf{a}$ in the domain of f is a

- global minimum if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of

f. In this case, $f(\mathbf{a})$ is the global minimum value, or absolute minimum value, of the function f.

- global maximum if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the global maximum value, or absolute maximum value, of the function f.
- local minimum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f. In this case, we say $f(\mathbf{a})$ is a local minimum value for the function f.
- local maximum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f. In this case, we say $f(\mathbf{a})$ is a local maximum value for the function f.

A local minimum or local maximum is also called a local extremum.

Remark

The graph of the function $-f$ is obtained by reflecting the graph of the function f in the output axis. This operation turns local and global maxima into local and global minima, and vice versa. So methods for finding

minima for a general function often give methods for finding maxima, which makes the two problems often equivalent.

Definition: Critical points

Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$.

Then we say $\mathbf{a}$ is a critical point if $\nabla f(\mathbf{a}) = \mathbf{0}$.

For a scalar-valued function of two variables, for example, $f(x, y)$, a critical point is a solution to the two equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$. Just as the derivative of a scalar-valued function of a single variable helps identify local minima and maxima, partial derivatives provide information about possible local minima/maxima.

Theorem

Suppose f is differentiable at an interior point $\mathbf{a}$ of its domain. If $\mathbf{a}$ is a local extremum of f, then

$$\nabla f(\mathbf{a}) = \mathbf{0}.$$

Why is this true?

Let $\mathbf{v}$ be a unit vector, and define

$$g(t) = f(\mathbf{a} + t\mathbf{v}).$$

Because $\mathbf{a}$ is an interior local extremum of f, the function g has a local extremum at $t = 0$. Therefore

$$g'(0) = 0.$$

By the theorem,

$$g'(0) = D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}.$$

This holds for every unit vector $\mathbf{v}$. If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, choose

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

Then

$$\nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| > 0,$$

a contradiction. Hence $\nabla f(\mathbf{a}) = \mathbf{0}$.

Warning: A necessary condition is not sufficient

For a function differentiable at an interior point $\mathbf{a}$, the theorem says that $\nabla f(\mathbf{a}) = \mathbf{0}$ is a *necessary* condition for a local extremum, but it is *not sufficient*. A critical point need not be a local minimum or a local maximum.

Example: Critical points of a difference of squares (U3-LO9)

Find the critical points of $f(x, y) = x^2 - y^2$.

We compute that $f_x(x, y) = 2x$ and $f_y(x, y) = -2y$. The only (x, y) such that $f_x(x, y) = 2x = 0$ and $f_y(x, y) = -2y = 0$ is $(0, 0)$. This is the only critical point.

Activity: Critical points of a quadratic function (U3-LO9)

Find the critical points of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y.$$

To find the critical points, we calculate the partial derivatives and set them equal to zero. We find that

$$f_x(x, y) = 2x - 2 \quad \text{and} \quad f_y(x, y) = 2y + 4.$$

The critical points are the pairs (x, y) that simultaneously solve the equations $2x - 2 = 0$ and $2y + 4 = 0$. So the only critical point is $(1, -2)$.

Activity: Critical points via row reduction (U3-LO9, U2-LO2)

Find all critical points of

$$f(x, y, z) = x^2 + xy + yz + z^2 - x - 2y - z.$$

Write the critical-point equations as an augmented matrix, use Gaussian elimination to obtain row echelon form, and finish by back-substitution.

The polynomial f is differentiable everywhere, and its gradient is

$$\nabla f(x, y, z) = \begin{bmatrix} 2x + y - 1 \\ x + z - 2 \\ y + 2z - 1 \end{bmatrix}.$$

Thus its critical points are exactly the solutions of

$$\begin{cases} 2x + y = 1, \\ x + z = 2, \\ y + 2z = 1. \end{cases}$$

Following The Gaussian Elimination Algorithm, we first swap rows to obtain a leading pivot of 1. We then eliminate below each pivot and normalize the final pivot:

$$\left[\begin{array}{ccc|c} 2 & 1 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 4 & 4 \end{array}\right]$$

$$\xrightarrow{R_3 \to \frac{1}{4} R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

The last row gives $z = 1$. Back-substitution into the second row, $y - 2z = -3$, and then the first row, $x + z = 2$, gives

$$y = -3 + 2z = -3 + 2(1) = -1,$$
$$x = 2 - z = 2 - 1 = 1.$$

There is a pivot in every variable column and no inconsistent row, so the system has a unique

solution. Therefore the only critical point of f is $(1, -1, 1)$.

The next critical-point system leads to a quadratic equation. Recall that, when $a \neq 0$, the real solutions of $at^2 + bt + c = 0$ are given by the *quadratic formula*

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

provided $b^2 - 4ac \geq 0$. If $b^2 - 4ac < 0$, there are no real solutions.

Activity: Critical points using the quadratic formula (U3-LO9)

Find all critical points of

$$f(x, y) = x^3 + 2xy + y^2 - 2x.$$

Use the quadratic formula to solve the resulting quadratic equation.

The partial derivatives are

$$f_x(x, y) = 3x^2 + 2y - 2, \qquad f_y(x, y) = 2x + 2y.$$

Setting them equal to zero gives

$$\begin{cases} 3x^2 + 2y - 2 = 0, \\ 2x + 2y = 0. \end{cases}$$

The second equation gives $y = -x$. Substituting into the first equation yields

$$3x^2 - 2x - 2 = 0.$$

With $a = 3$, $b = -2$, and $c = -2$, the quadratic formula gives

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(3)(-2)}}{2(3)} = \frac{2 \pm \sqrt{28}}{6} = \frac{1 \pm \sqrt{7}}{3}.$$

For each root, use $y = -x$. Thus the two critical points are

$$\left(\frac{1+\sqrt{7}}{3}, \frac{-1-\sqrt{7}}{3}\right) \quad \text{and} \quad \left(\frac{1-\sqrt{7}}{3}, \frac{-1+\sqrt{7}}{3}\right).$$

These are all the critical points because every solution of the system must satisfy $y = -x$ and one of the two quadratic roots above.

When a critical-point equation factors as $(x - a)(y - b) = 0$, at least one factor must be zero: $x = a$ or $y = b$, possibly both. Treat these as two cases, substitute each into the remaining equation, and combine the

solutions, listing any repeated point only once. Dividing by either factor without considering when it is zero can discard valid solutions.

Activity: Critical points by splitting into cases (U3-LO9)

Find all critical points of

$$f(x, y) = (x - 1)^2(y - 2) + y^2 - 8y.$$

The partial derivatives are

$$f_x(x, y) = 2(x - 1)(y - 2),$$
$$f_y(x, y) = (x - 1)^2 + 2y - 8.$$

Setting them equal to zero and dividing the first equation by 2 gives

$$\begin{cases} (x - 1)(y - 2) = 0, \\ (x - 1)^2 + 2y - 8 = 0. \end{cases}$$

The first equation gives two cases, and in each case the second equation must also hold.

- If $x = 1$, the second equation becomes $2y - 8 = 0$, so $y = 4$. This gives the critical point $(1, 4)$.

- If $y = 2$, the second equation becomes

$$(x-1)^2 - 4 = 0.$$

Thus $x - 1 = \pm 2$, giving $x = -1$ or $x = 3$. This gives the critical points $(-1, 2)$ and $(3, 2)$.

Therefore all critical points are

$$(1, 4), \qquad (-1, 2), \qquad (3, 2).$$

Each point satisfies both equations, and the two cases exhaust the solutions of the first equation.

Warning: Critical points are candidates

The equation $\nabla f = 0$ finds candidates for local extrema. It does not by itself find boundary extrema, constrained extrema, or absolute extrema. Always combine critical-point information with the domain and boundary conditions.

Activity: Redundant features give a flat direction of the loss (U3-LO6, U2-LO6)

Recall Anatomy of a linear map's redundant-feature example in Two coefficient choices, same predictions (U2-LO1, U2-LO6). Add a third feature to our data by summing the original two feature columns:

Activity: Redundant features give a flat direction of the loss (U3-LO6, U2-LO6) (continued)

$$\widetilde{X} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

There are now three weights $\theta = (\theta_1, \theta_2, \theta_3)$, with loss

$$\widetilde{\mathscr{L}}(\theta) = \|\widetilde{X}\theta - \mathbf{y}\|^2.$$

The predictions depend only on the two combinations $w_1 = \theta_1 + \theta_3$ and $w_2 = \theta_2 + \theta_3$, so $\widetilde{\mathscr{L}}(\theta) = \mathscr{L}(\theta_1 + \theta_3, \theta_2 + \theta_3)$.

1. Verify that $\mathbf{z} = (-1, -1, 1)$ is in the null space of $\widetilde{X}$.
2. Show that $\widetilde{\mathscr{L}}(\theta + t\mathbf{z}) = \widetilde{\mathscr{L}}(\theta)$ for every real t. Evaluate this constant loss along the line through $\theta_0 = (1, 1, 0)$.
3. What is $\frac{d}{dt}\widetilde{\mathscr{L}}(\theta + t\mathbf{z})$? If a minimizing weight vector is found, can it be the unique minimizing weight vector?

1. Direct multiplication verifies the given direction:

$$\widetilde{X}\mathbf{z} = \begin{bmatrix} -1 + 1 \\ -1 + 1 \\ -1 - 1 + 2 \end{bmatrix} = \mathbf{0}.$$

Decreasing each original weight by t and increasing the redundant-feature weight by t

leaves both effective weights unchanged.

2. Unit 2's linearity identity gives

$$\widetilde{X}(\theta + t\mathbf{z}) - \mathbf{y} = \widetilde{X}\theta - \mathbf{y} + t\widetilde{X}\mathbf{z} = \widetilde{X}\theta - \mathbf{y}.$$

   Taking squared lengths proves equality of the losses for every t. At θ_0, the predictions are $(1, 1, 2)$ and the loss is 1. Every point $(1 - t, 1 - t, t)$ gives these same predictions and the same loss.

3. The loss along the line is constant, so its derivative is zero for every t. By the chain rule, $\nabla\widetilde{\mathscr{L}}(\theta)^T\mathbf{z} = 0$. If θ_* minimizes the loss, all distinct vectors $\theta_* + t\mathbf{z}$ minimize it as well. The data therefore cannot identify these three weights uniquely.

Note: Exact invariance and first-order change

For a linear prediction map $\theta \mapsto \widetilde{X}\theta$, any direction $\mathbf{z}$ satisfying $\widetilde{X}\mathbf{z} = \mathbf{0}$ leaves the predictions, and therefore the loss, unchanged along the entire line $\theta + t\mathbf{z}$. This is the exact invariance illustrated in Redundant features

give a flat direction of the loss (U3-LO6, U2-LO6).
For a differentiable nonlinear map F, the condition $J_F(\mathbf{a})\mathbf{z} = \mathbf{0}$ guarantees only zero first-order change at $\mathbf{a}$. The output $F(\mathbf{a} + t\mathbf{z})$ may still vary with t, as in Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3). Also, equal loss values do not necessarily mean equal predictions: different prediction vectors can have the same squared error, as From predictions to a scalar error function (U3-LO1, U2-LO1) illustrates.

Gradient descent

Example: Critical-point equations in three variables (U3-LO9)
Find the system of equations satisfied by the critical points of the function

$$f(x, y, z) = x^2 e^{yz} + y \sin(xz) + z \cos(xy).$$

We calculate that

$$f_x(x, y, z) = 2xe^{yz} + yz \cos(xz) - yz \sin(xy),$$

that

$$f_y(x, y, z) = x^2 z e^{yz} + \sin(xz) - xz \sin(xy)$$

and that

$$f_z(x, y, z) = x^2 y e^{yz} + xy \cos(xz) + \cos(xy).$$

So the critical points of f are the solutions to the system of equations

$$\begin{cases} 2xe^{yz} + yz\cos(xz) - yz\sin(xy) = 0, \\ x^2ze^{yz} + \sin(xz) - xz\sin(xy) = 0, \\ x^2ye^{yz} + xy\cos(xz) + \cos(xy) = 0. \end{cases}$$

Using the calculus we have developed, it is usually straightforward to write the equations that critical points must satisfy. Critical-point equations in three variables (U3-LO9) also shows the problem: those equations may be nonlinear and difficult or impossible to solve exactly. This is different from the linear systems in Unit 2, where Gaussian elimination gave a systematic exact method.

When the critical-point equations cannot be solved directly, one often uses a numerical method. By the theorem, the negative gradient gives the direction of steepest local decrease.

At a current point $\mathbf{x}_k$, choose

$$\mathbf{h} = -\alpha\nabla f(\mathbf{x}_k), \qquad \alpha > 0.$$

By Linear Approximation,

$$\begin{aligned} f(\mathbf{x}_k + \mathbf{h}) - f(\mathbf{x}_k) &\approx \nabla f(\mathbf{x}_k)^T\mathbf{h} \\ &= -\alpha\|\nabla f(\mathbf{x}_k)\|^2. \end{aligned}$$

Unless the gradient is zero, the local model predicts a decrease. This is a local prediction, not a convergence theorem. It motivates the gradient-descent update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The gradient-descent algorithm repeats this local step.

1. Start at an initial point $\mathbf{x}_0$.
2. Compute the gradient $\nabla f(\mathbf{x}_0)$.
3. Move in the direction of the negative gradient: set $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0)$, where $\alpha_0 > 0$ is a step size (or learning rate) parameter.
4. Repeat steps 2-3 with step sizes $\alpha_1, \alpha_2, \ldots$ to successively define points $\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1 \nabla f(\mathbf{x}_1)$, $\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2)$, and so on, with the hope that these points converge to a local minimum.

In data and modeling problems, an objective function being minimized is often called a loss.
Intuitively, gradient descent works because each successive value $\mathbf{x}_k$ is obtained by shifting the previous value $\mathbf{x}_{k-1}$ in the direction in which the function f decreases the fastest. Gradient descent may converge to a local minimum under additional assumptions on the function and the learning rates. The examples below illustrate possible behavior; they do not prove convergence.

Activity: Three gradient-descent steps (U3-LO9)

Find an approximate minimum of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$$

using gradient descent. Use the starting point $\mathbf{x}_0 = (0, 0)$, a constant step size of 0.1, and stop after computing $\mathbf{x}_3$. You may use a calculator to perform these computations.

First, we calculate the gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x - 3y + 2 \\ -3x + 6y + 5 \end{bmatrix}$$

Starting at $\mathbf{x}_0 = (0, 0)$, we successively compute that

$$\nabla f(0, 0) = \begin{bmatrix} 2 \\ 5 \end{bmatrix},$$

so

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix}.$$

We then compute that

$$\begin{aligned} &\nabla f(-0.2, -0.5) \\ &= \begin{bmatrix} -0.4 - (-1.5) + 2 \\ -3(-0.2) + 6(-0.5) + 5 \end{bmatrix} \\ &= \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix}, \end{aligned}$$

so

$$\mathbf{x}_2 = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix} = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix}.$$

Finally, we compute that

$$\begin{aligned} &\nabla f(-0.51, -0.76) \\ &= \begin{bmatrix} 2(-0.51) - 3(-0.76) + 2 \\ -3(-0.51) + 6(-0.76) + 5 \end{bmatrix} \\ &= \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix}, \end{aligned}$$

so

$$\mathbf{x}_3 = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix} - 0.1 \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix} = \begin{bmatrix} -0.836 \\ -0.957 \end{bmatrix}.$$

In Revisiting the gradient-descent quadratic (U5-LO5, U3-LO9), we compute exactly that the unique critical point is

$$\left(-9, -\frac{16}{3}\right)$$

and classify it after developing second-order tools. Thus $\mathbf{x}_3$ is not yet a good approxima-

tion. If we continue the iteration, the points $\mathbf{x}_k$ for, say, $k \geq 1000$ are very close to that critical point.

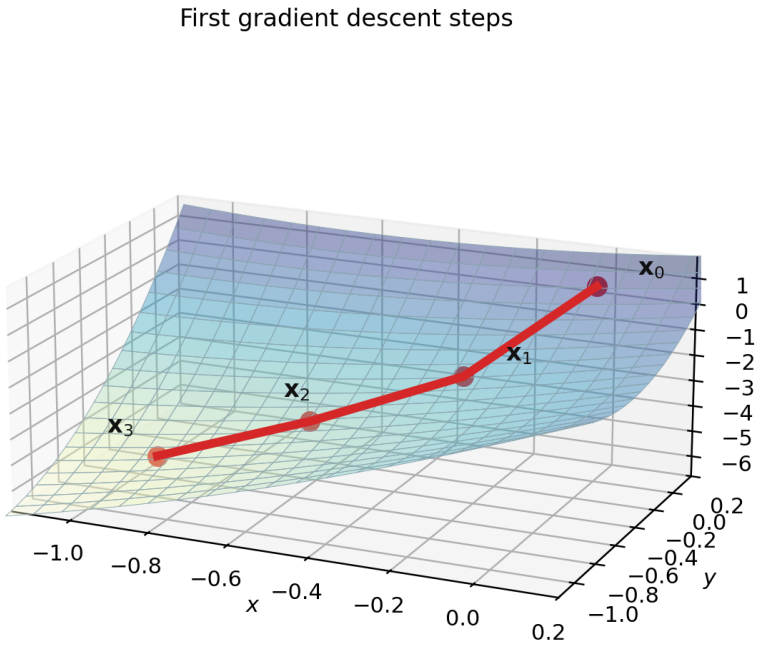


Figure. Surface plot showing the first gradient descent steps from the origin.

Activity: From a symbolic gradient to a descent step (U3-LO9)

Use SymPy to check the first step of the preceding calculation for $f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$. We construct a gradient as in Building a gradient and Hessian in SymPy (U3-LO3), evaluate it at the starting point, and apply the gradient-descent update. The learning rate `sp.Rational(1, 10)` represents 0.1 exactly.

```
import sympy as sp
```

Activity: From a symbolic gradient to a descent step (U3-LO9) (continued)

```
x, y = sp.symbols("x y")
f = x**2 - 3*x*y + 3*y**2 + 5*y + 2*x
grad_f = sp.Matrix([sp.diff(f, x), sp.diff(f, y)])

p0 = sp.Matrix([0, 0])
g0 = grad_f.subs({x: p0[0], y: p0[1]})
alpha = sp.Rational(1, 10)
p1 = p0 - alpha*g0
f0 = f.subs({x: p0[0], y: p0[1]})
f1 = f.subs({x: p1[0], y: p1[1]})

print(grad_f)
print(g0)
print(p1)
print(f0, f1)
print((g0.T * (p1 - p0))[0], f1 - f0)
```

Output:

```
Matrix([[2*x - 3*y + 2], [-3*x + 6*y + 5]])
Matrix([[2], [5]])
Matrix([[-1/5], [-1/2]])
0 -241/100
-29/10 -241/100
```

In the last line, `g0.T * (p1 - p0)` is a 1×1 matrix. The index `[0]` extracts its scalar entry, the local linear prediction for the change in f.

1. What is the difference between `grad_f` and `g0`? What shapes must `p0`, `g0`, and `p1` have for the update to make sense?
2. Write the numerical vector update that produces `p1`. Did the function value decrease? Compare the predicted change with the actual change, using the scalar linear approximation from Linear Approximation.

Activity: From a symbolic gradient to a descent step (U3-LO9) (continued)

3. Predict `grad_f.subs({x: p1[0], y: p1[1]})`. Is `p1` a critical point? Explain which symbolic expression can be reused at the next step and which evaluated vector must be recomputed.
4. If `alpha` is changed to `1` and the subsequent lines are rerun, what are `p1` and `f1`? What does this comparison show about the learning rate?

1. The matrix `grad_f` contains expressions for the two partial derivatives as functions of x, y; `g0` is their value at $(0, 0)$. Each point and evaluated gradient is a 2×1 column, so the scalar multiple `alpha*g0` can be subtracted from `p0`.
2. The update is

$$\mathbf{p}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \frac{1}{10}\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -1/5 \\ -1/2 \end{bmatrix}.$$

The function value decreases from 0 to $-241/100 = -2.41$. The local linear approximation predicts

$$\nabla f(\mathbf{p}_0)^T(\mathbf{p}_1 - \mathbf{p}_0) = -\frac{1}{10}(2^2 + 5^2) = -\frac{29}{10}.$$

The actual change is $-241/100$. Both are negative, but the actual decrease is smaller than predicted because the linear approximation omits nonlinear terms.

3. Evaluating at the new point gives

$$\nabla f(-1/5, -1/2) = \begin{bmatrix} 31/10 \\ 13/5 \end{bmatrix},$$

so $\mathbf{p}_1$ is not a critical point. The symbolic expression `grad_f` can be reused without differentiating again. Its value must be recomputed at the new point; reusing the old vector `g0` would not implement gradient descent at that point.

4. With $\alpha = 1$, the new point is $(-2, -5)$ and $f(-2, -5) = 20$, an increase from 0. The local model predicts a change of -29, but this larger step goes too far for that prediction

to ensure a decrease. A negative-gradient direction does not guarantee a decrease for every positive learning rate.

Remark

The iteration is cumbersome by hand but easy to repeat on a computer when the gradient is available.

Activity: Improving a model's predictions with one step (U3-LO8, U3-LO9)

Return to the three observations from From predictions to a scalar error function (U3-LO1, U2-LO1) and the loss

$$\mathscr{L}(w_1, w_2) = (w_1 - 1)^2 + (w_2 - 2)^2 + (w_1 + w_2 - 2)^2.$$

At the starting weights $\mathbf{w}_0 = (1, 1)$, the loss is 1 and the gradient is $(0, -2)$. In this model, a training step adjusts the weights while the features X and observations $\mathbf{y}$ remain fixed.

1. Find the unit direction of greatest decrease at $\mathbf{w}_0$ and its directional derivative. Take one gradient-descent step with learning rate $\alpha = 1/10$.
2. Compute the new prediction vector, residual, and loss. Compare the actual change with the first-order prediction $-\alpha\|\nabla\mathscr{L}(\mathbf{w}_0)\|^2$.
3. Evaluate the gradient at the new weights. Has the step reached a critical point?
4. Starting again at $\mathbf{w}_0$, what happens if $\alpha = 1$? Explain why the same descent direction can give an increase in loss.

1. The negative gradient is $(0, 2)$, so its unit direction is $(0, 1)$. The directional derivative is $(0, -2) \cdot (0, 1) = -2$. Gradient descent uses the unnormalized gradient:

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6/5 \end{bmatrix}.$$

2. The new predictions and residuals are

$$X\mathbf{w}_1 = \begin{bmatrix} 1 \\ 6/5 \\ 11/5 \end{bmatrix}, \qquad R(\mathbf{w}_1) = \begin{bmatrix} 0 \\ -4/5 \\ 1/5 \end{bmatrix}.$$

The loss is $16/25 + 1/25 = 17/25 = 0.68$, down from 1. The predicted change is $-(1/10)(4) = -2/5$, while the actual change is $-8/25$. Their difference, $2/25$, is the squared length of $X(\mathbf{w}_1 - \mathbf{w}_0) = (0, 1/5, 1/5)$, as in Predicting a change in fitting error (U3-LO4).

3. The gradient formula from Differentiating a loss through its predictions (U3-LO6) gives

$$\nabla \mathscr{L}(\mathbf{w}_1) = 2X^T \begin{bmatrix} 0 \\ -4/5 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 2/5 \\ -6/5 \end{bmatrix}.$$

   It is nonzero, so the new weights are not a critical point. The step improves the fit without finishing the minimization.

4. With $\alpha = 1$, the new weights are $(1, 3)$, the predictions are $(1, 3, 4)$, and the residual is $(0, 1, 2)$. The loss becomes 5, which is larger than the starting value. The negative gradient specifies a direction of local decrease; the learning rate determines how far we travel. A step that is too large can increase the loss, just as in From a symbolic gradient to a descent step (U3-LO9).

These examples use derivatives to improve a fit and interpret directions that leave it unchanged. Unit 4 develops the general least-squares

method for finding a best-fitting prediction vector and its geometric interpretation.